Real-World Examples of Cybersecurity Threats

Har threat ki aik real-world example dekho, taake notes mein easily likh sako aur concept bhi clear ho jaye.

1. Malware Threats

1. Virus: A student downloads an infected USB file. When the file runs, the virus infects other files on the computer.

2. Worm: A worm spreads automatically from one computer to other vulnerable computers connected to the same network.

3. Trojan Horse: A user downloads a fake video player that secretly installs malicious software on their computer.

4. Ransomware: A hospital's computers are infected with ransomware, and attackers demand money to restore access to patient files.

5. Spyware: A malicious app secretly collects a user's browsing activity and personal information.

6. Adware: A user installs a free application that continuously displays unwanted advertisements.

7. Rootkit: An attacker installs a rootkit to hide malicious processes and maintain unauthorized access to a compromised computer.

8. Keylogger: A user types their banking password on an infected computer, and a keylogger records the keystrokes.

2. Phishing Attacks

9. Email Phishing: A student receives a fake email claiming their university account will be blocked unless they verify their password.

10. Spear Phishing: An attacker sends a customized email to a university employee using their name and department to trick them into opening a malicious attachment.

11. Whaling: An attacker sends a fake financial approval request to a company's CEO to steal money or confidential information.

12. Smishing: A person receives a fake SMS claiming that their parcel is delayed and asking them to click a fraudulent link.

13. Vishing: A scammer calls someone pretending to be a bank representative and asks for their OTP.

3. Social Engineering

14. Pretexting: An attacker pretends to be an IT technician and asks an employee for login credentials to supposedly fix a computer problem.

15. Baiting: Someone leaves a USB drive labelled “Confidential Salary Data” near an office computer, hoping an employee will plug it in.

16. Quid pro quo: A fake technical support agent offers free computer repair in exchange for the user's password.

17. Tailgating: An unauthorized person follows an employee through a secure office door without using an access card.

18. Shoulder Surfing: Someone watches a student enter their ATM PIN and later attempts to misuse it.

4. Password Attacks

19. Brute-Force Attack: An attacker uses automated software to try many password combinations against an account.

20. Dictionary Attack: An attacker tries common passwords such as password123 and qwerty123 to access an account.

21. Credential Stuffing: An attacker uses passwords leaked from one website to attempt access to the victim's accounts on other websites.

22. Password Spraying: An attacker tries a common password against many employee accounts rather than trying many passwords against one account.

23. Keylogging: An attacker captures a user's email password through a malicious program that records keystrokes.

5. DoS and DDoS Attacks

24. DoS Attack: An attacker floods a small business website with requests from one system, making the website unavailable to customers.

25. DDoS Attack: An attacker uses thousands of compromised devices to flood an online shopping website, preventing legitimate customers from accessing it.

6. Man-in-the-Middle (MitM)

26. MitM Attack: A student connects to a malicious public Wi-Fi hotspot. The attacker intercepts unencrypted information sent by the student's device.

7. Injection Attacks

27. SQL Injection: An attacker exploits an insecure login form to manipulate a database query and potentially access unauthorized records.

28. Command Injection: An attacker exploits a vulnerable web application to make the server execute unintended operating-system commands.

29. Cross-Site Scripting (XSS): An attacker injects a malicious script into a vulnerable comment section, causing it to run in visitors' browsers.

8. Spoofing Attacks

30. IP Spoofing: An attacker forges the source IP address of network packets to make them appear to originate from another system.

31. Email Spoofing: An employee receives an email that appears to come from the company's manager, requesting an urgent payment.

32. DNS Spoofing: A user enters a bank's website address but is redirected to a fraudulent website because of manipulated DNS information.

33. Website Spoofing: A scammer creates a fake shopping website resembling a legitimate store to steal customers' login and payment information.

9. Insider Threats

34. Malicious Insider: An employee deliberately copies confidential customer information and shares it with an unauthorized person.

35. Negligent Insider: An employee accidentally sends a confidential company file to the wrong email recipient.

36. Compromised Insider Account: An attacker steals an employee's password and uses the employee's account to access confidential company documents.

10. Zero-Day Threat

37. Zero-Day Attack: Attackers discover and exploit an unknown vulnerability in a company's software before the vendor releases a security patch.

11. Advanced Persistent Threat (APT)

38. APT: A cyber-espionage group secretly accesses a government network and remains there for months, collecting confidential information.

12. Botnet Threat

39. Botnet: Attackers infect thousands of internet-connected devices, such as computers and cameras, and control them together to launch a DDoS attack.

13. Physical Threats

40. Laptop Theft: Someone steals an employee's laptop containing confidential company files.

41. Unauthorized Entry: An intruder enters a server room and connects an unauthorized device to the company's network.

42. Equipment Damage: A fire damages a company's servers, disrupting its services and potentially causing data loss.

14. Supply Chain Attack

43. Supply Chain Attack: Attackers compromise a trusted software vendor's update system, causing customers who install the malicious update to become vulnerable.

15. Cloud Security Threats

44. Misconfigured Cloud Storage: A company accidentally makes its cloud storage publicly accessible, exposing confidential customer documents.

45. Stolen Cloud Credentials: An attacker steals an administrator's cloud password and accesses sensitive company data.

46. Insecure API: A company's application programming interface (API) fails to check user permissions properly, allowing one customer to access another customer's records.

47. Excessive Permissions: A cloud employee has administrator-level access despite needing access only to a small set of files.

48. Cloud Data Leakage: An employee accidentally uploads a confidential company database to a publicly accessible cloud folder.